Year 2 Statistics and Mechanics

Chapter 7

Applications of forces

Practice Booklet

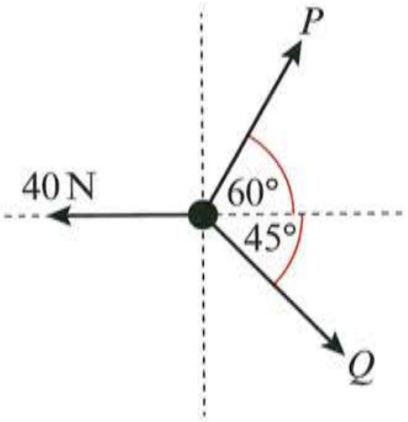
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7.1 Static particles

Practice Book, printed page 56.

1. A particle is held in static equilibrium by the forces shown in the diagram.

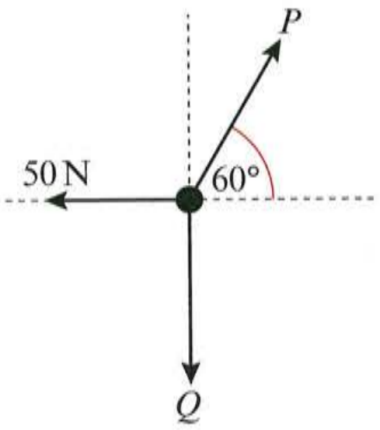


a. Resolve the components in the x-direction.
b. Resolve the components in the y-direction.
c. Find the magnitudes of P and Q.

Hint

Solve your equations from parts **a** and **b** simultaneously.

2. A particle is held in equilibrium by the forces shown in the diagram.



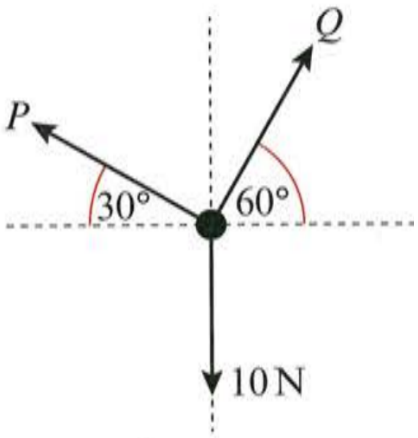
a. Draw a triangle of forces diagram.

b. Find the magnitudes of the forces P and Q.

Hint

Use trigonometry to find the values of P and Q.

3. A particle is held in equilibrium by the forces shown in the diagram.

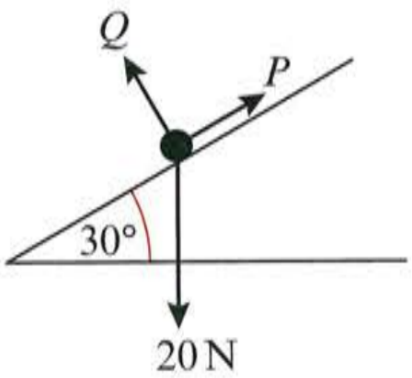


Find the magnitudes of the forces P and Q.

Hint

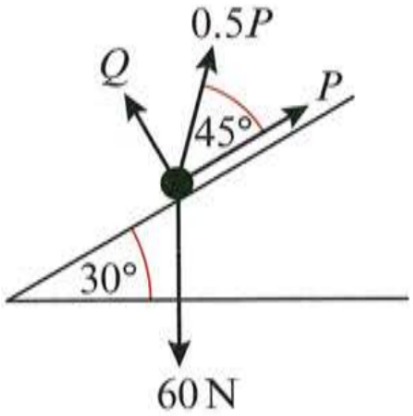
Resolve horizontally and vertically then solve the resulting equations simultaneously.
← Section 5.1

4. A particle is held in equilibrium on a slope by the forces shown in the diagram.



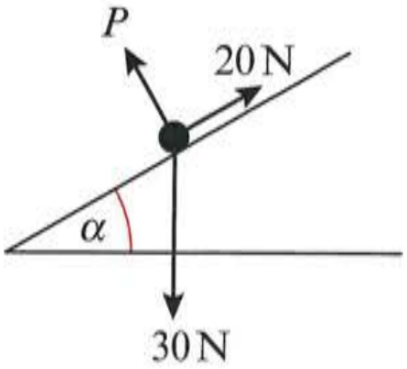
Find the magnitudes of the forces P and Q.

5. A particle is held in equilibrium on a slope by the forces shown in the diagram.



Find the magnitudes of the forces P and Q.

6. A particle is held in equilibrium on a slope by the forces shown in the diagram.



Given that the slope is angled at α to the horizontal, find:

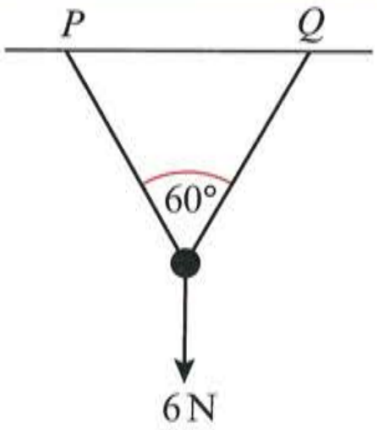
a. the size of angle α

b. the magnitude of P.

7.2 Modelling with statics

Practice Book, printed page 57.

1. A smooth bead of weight 6 N is threaded on a light inextensible string that is attached to points P and Q as shown.

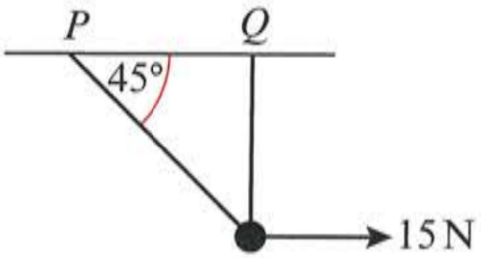


Work out the tension in the string.

Hint

Because the bead is smooth, the tension in the string is the same on either side of the bead.

2. A smooth bead is threaded on a light inextensible string that is attached to points P and Q as shown. The bead is held in equilibrium vertically below Q by a horizontal force of 15 N.



a. Find the tension in the string.

b. Work out the weight of the bead.

c. How have you used the fact that the bead is smooth in your calculations?

Hint

Copy the diagram and show all the forces acting on the bead.

3. A box of mass 5 kg rests on a smooth plane that is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{5}{12}$. A force of P N acts in a direction parallel to the plane and holds the box in equilibrium.

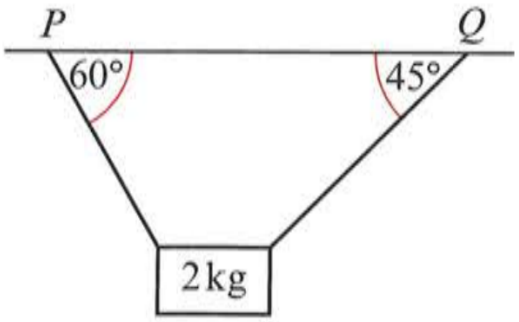
a. Find the normal reaction between the box and the plane.

b. Work out the value of P.

Hint

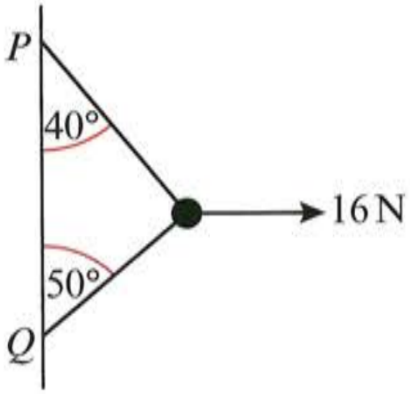
Resolve parallel and perpendicular to the slope to find the normal reaction.

4. E A sign of mass 2 kg hangs in equilibrium as shown on the diagram.



It is attached to the points P and Q by light inextensible strings.
Work out the tension in each of the strings. (5 marks)

5. E A smooth bead is threaded on a light inextensible string that is attached to a vertical wall at the points P and Q as shown in the diagram. The bead is held in equilibrium by a horizontal force of magnitude 16 N.



a. Find the tension in the string. (2 marks)

b. Work out the mass of the bead. (2 marks)

c. State how the tension in the string would be affected if the string were not modelled as being light. Justify your answer. (2 marks)

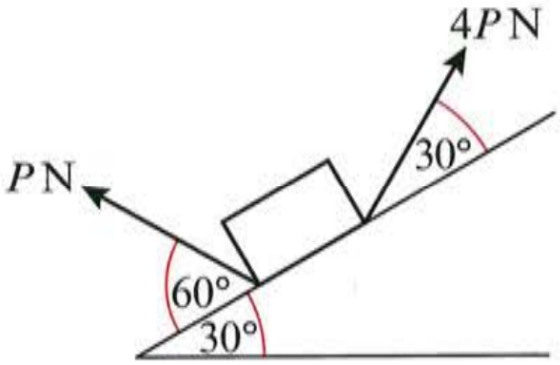
6. E/P A particle of mass 0.2 kg rests on a smooth plane that is inclined at an angle α to the horizontal. The particle is held in equilibrium by a horizontal force of magnitude 2 N.

a. Find the size of angle α (3 marks)

b. State how your answer to part **a** would change if the particle was heavier. (1 mark)

7. E/P A box of mass m kg rests on a smooth plane that is inclined at an angle α to the horizontal. A force of $2g$ N acts in a direction parallel to and up the plane and holds the box in equilibrium.
Show that $\alpha = \arcsin\left(\frac{2}{m}\right)$ (2 marks)

8. E/P A box of weight 20 N rests on a smooth slope that is inclined at an angle of 30° to the horizontal. The box is held in equilibrium by a force of magnitude $4P$ N that acts at an angle of 30° to the slope and a second force of magnitude P N that acts at an angle of 60° to the slope, as shown in the diagram.



Show that the normal reaction between the box and the slope is 7.65 N to 3 significant figures. (8 marks)

7.3 Friction and static particles

Practice Book, printed page 58.

1. A box of mass 2 kg rests on a rough horizontal plane. The coefficient of friction between the box and the plane is $\mu = 0.2$. A force P acts on the box in a direction parallel to the plane.

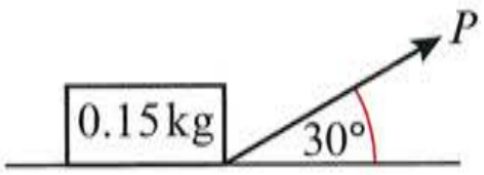
a. Draw a diagram to show the forces acting on the box.

b. Given that the box remains at rest, find the maximum magnitude of P.

Hint

Use $F \leq \mu R$ to work out the magnitude of the frictional force.
← Section 5.3

2. A box of mass 0.15 kg rests on a rough horizontal plane. The coefficient of friction between the box and the plane is 0.25. A force P acts on the box at an angle of 30° to the plane.



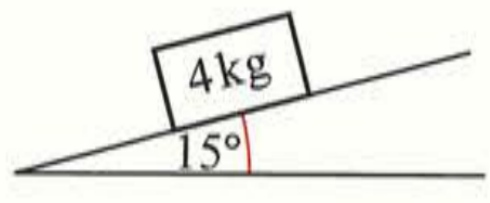
a. Complete the diagram showing all the forces acting on the box.

b. Given that the box remains at rest, find the maximum magnitude of P.

Hint

First resolve vertically to write the normal reaction in terms of P.

3. A box of mass 4 kg rests on a rough plane that is inclined at 15° to the horizontal. The coefficient of friction between the box and the plane is μ.



Given that the box is in limiting equilibrium, find:

a. the normal reaction between the box and the plane

b. the value of μ.

Hint

When the box is in limiting equilibrium, it is on the point of slipping, and $F = \mu R$.

4. E/P A box of mass 2.5 kg rests on a rough plane that is inclined at 18° to the horizontal. The coefficient of friction between the box and the plane is 0.3. A force of P N acts on the box parallel to and up the plane. Given that the box is in limiting equilibrium, find:

a. the normal reaction between the box and the plane (2 marks)

b. the size of the maximum force that can be applied to the box without causing it to move. (3 marks)

5. E/P A block of mass m kg rests on a rough plane that is inclined at 30° to the horizontal. The coefficient of friction between the box and the plane is 0.4. A force of magnitude 5 N acts on the block parallel to and up the plane.
Given that the box is in limiting equilibrium, find the two possible values of m. (5 marks)

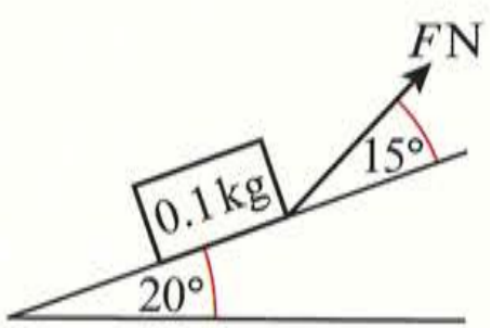
6. E/P A particle of mass 1 kg rests in limiting equilibrium on a rough slope that is inclined at an angle of α to the horizontal. A force of magnitude g N acts on the particle parallel to and down the slope. Given that the particle remains in equilibrium, show that the minimum value of the coefficient of friction between the box and the plane is $\mu = \dfrac{1 + \sin\alpha}{\cos\alpha}$ (4 marks)

7. E/P A boat of mass 250 kg is held at rest on a slipway that is angled at 25° to the horizontal by a rope that acts parallel to and up the slope. The tension in the rope is P N. The coefficient of friction between the slipway and the boat is 0.15. The boat can be modelled as a particle and the rope as a light inextensible string.

a. Given that the boat is on the point of slipping down the slope, work out the value of P. (4 marks)

b. Comment on the validity of modelling the boat as a particle in this situation. (1 mark)

8. E/P A block of mass 0.1 kg rests on a slope that is inclined at 20° to the horizontal. The coefficient of friction between the block and the slope is 0.25. The block is held in equilibrium by a force of magnitude F N that acts up the slope and at an angle of 15° to the slope.

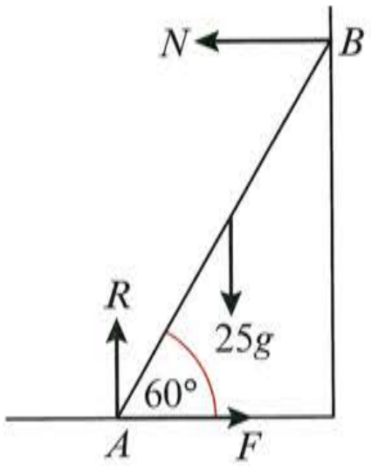


By modelling the block as a particle, find the range of possible values of F. (8 marks)

7.4 Static rigid bodies

Practice Book, printed page 60.

1. A uniform ladder AB, of mass 25 kg and length 6 m, rests with end A on rough horizontal ground and end B on a smooth vertical wall. The angle between the ladder and the ground is 60°.



Given that the ladder is in limiting equilibrium, find:

a. the normal reaction, N, between the ladder and the wall in terms of F

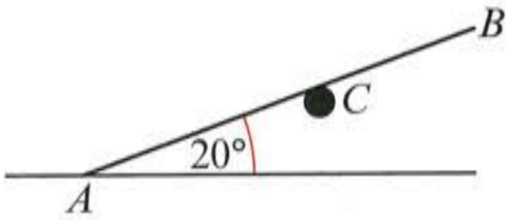
b. the normal reaction between the ladder and the ground.

c. By taking moments about the base of the ladder, work out the magnitude of N.

Hint

For part **a**, resolve horizontally and for part **b** resolve vertically.

2. A uniform rod AB, of mass 20 kg and length 5 m, rests with end A on rough horizontal ground. The rod is supported by a smooth peg at the point C where $AC = 3.5$ m. The angle between the rod and the ground is 20°. The rod is in limiting equilibrium.



a. Copy and complete the diagram to show the forces acting on the rod.

b. Work out the normal reaction between the rod and the peg.

c. Find the coefficient of friction between the rod and the ground.

d. State how you have used the fact that the peg is smooth in your calculations.

Hint

Take moments about A to find the normal reaction at the peg.

3. A uniform ladder AB, of mass 10 kg and length 3 m, rests with end A on rough horizontal ground and end B on a smooth vertical wall.
The angle between the ladder and the ground is 60°. The ladder is in limiting equilibrium.

a. Draw a diagram to show the forces acting on the ladder.

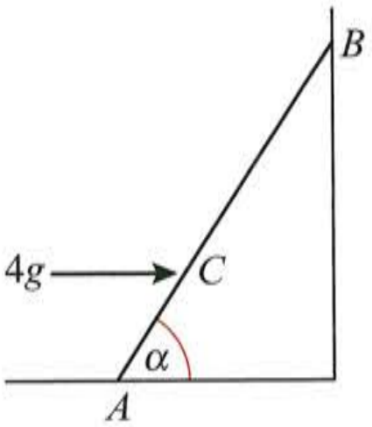
b. Work out the magnitude of the normal reaction between the ladder and the wall.

c. Find the coefficient of friction between the ladder and the ground.

Hint

Resolve horizontally and vertically then take moments about A.

4. E/P A uniform ladder AB, of length 4 m and mass 12 kg, rests with end A on smooth horizontal ground and the end B against a smooth vertical wall. The ladder is kept in equilibrium by a horizontal force of magnitude $4g$ N acting at a point C on the ladder, where $AC = 1$ m.



The angle between the ladder and the floor is α.

a. By modelling the ladder as a uniform rod, find the size of angle α. (4 marks)

b. Comment on the modelling assumption that the ground and the wall are smooth. (1 mark)

The ground and the wall are now modelled as rough surfaces, and the rod is assumed to be in limiting equilibrium.

c. Without further calculation, state whether the size of angle α will increase or decrease. (1 mark)

5. E/P A uniform ladder AB, of length 6 m and mass 24 kg, rests in equilibrium with end A on rough horizontal ground and end B against a smooth vertical wall. The ladder is inclined at an angle α to the horizontal. When a builder of mass 72 kg stands at a point C on the ladder, where $AC = 4$ m, the ladder is at the point of slipping. Given that the coefficient of friction between the ladder and the ground is 0.3, find:

a. the size of the frictional force of the ground acting on the ladder (3 marks)

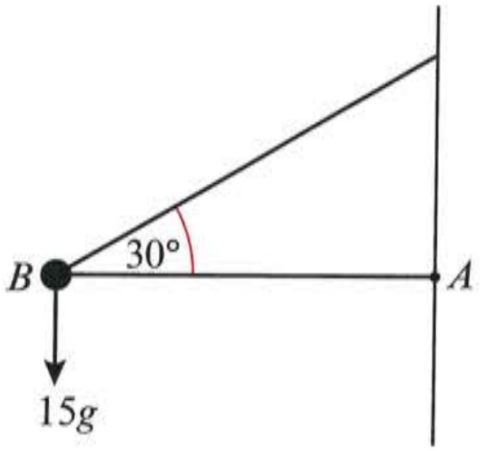
b. the size of angle α. (6 marks)

6. E/P A uniform plank AB, of weight W N and length 6 metres, rests on a smooth peg at a point C. The plank rests in limiting equilibrium with end A on rough horizontal ground, where the coefficient of friction between the plank and the ground is 0.8. The plank is inclined at an angle α to the horizontal, where $\tan\alpha = 0.75$. Given that the magnitude of the normal reaction of the ground on the plank at A is 80 N, find:

a. the magnitude of the normal reaction between the plank and the peg (5 marks)

b. the value of W. (3 marks)

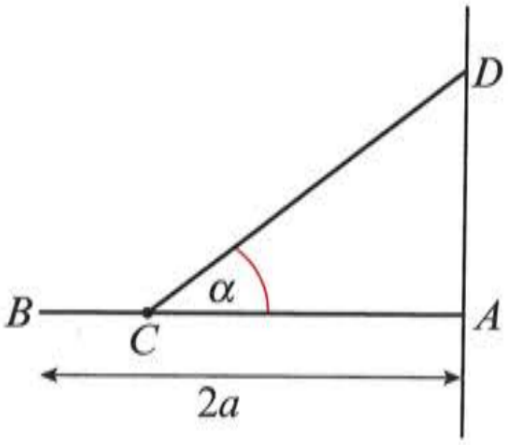
7. E/P A uniform rod AB, of length 4 m and mass 25 kg, is smoothly hinged at the point A, which lies on a vertical wall. A particle of mass 15 kg is suspended from B. The rod is kept in a horizontal position by a light inextensible string, which makes an angle of 30° with the rod, as shown in the diagram.



a. Work out the tension in the string. (2 marks)

b. Calculate the magnitude and direction of the reaction at the hinge. (8 marks)

8. E/P A uniform rod AB, of mass M and length $2a$, rests with its end A against a rough vertical wall. The rod is held in a horizontal position by a rope. One end of the rope is attached to the rod at a point C that lies $\frac{1}{2}a$ from B, and the other end is attached to the wall at a point D that lies vertically above A. The coefficient of friction between the wall and the rod is 0.4 and the angle between the rope and the rod is α.



a. Show that the tension in the rope is $T = \dfrac{2Mg}{3\sin\alpha}$ (3 marks)

b. Given that the rod is in limiting equilibrium, show that the magnitude of the horizontal component of the force exerted on the rod by the wall at A is $\dfrac{2Mg}{3\tan\alpha}$ and find the magnitude of the vertical component of this force. (3 marks)

c. Find the size of angle α. (3 marks)

d. Explain how your answer to part **c** would change if the coefficient of friction between the rod and the wall were greater than 0.4. (1 mark)

7.5 Dynamics and inclined planes

Practice Book, printed page 62.

1. A box of mass 5 kg is pushed up a rough surface, which is inclined at 30° to the horizontal, by a force of 120 N that acts in the direction parallel to and up the surface. The coefficient of friction between the box and the surface is 0.25.

a. Draw a force diagram to represent this situation.

b. Find the normal reaction between the box and the surface.

c. Work out the acceleration of the box.

Hint

Use $F = ma$ to set up an equation of motion for the box.

2. A block of mass 1 kg is held at rest on a rough surface that is inclined at 60° to the horizontal. The coefficient of friction between the block and the surface is 0.4.
The block is released from rest.

a. Calculate the frictional force between the block and the surface, stating in your answer in which direction it acts.

b. Show that the block will begin to slide down the slope.

c. Work out the acceleration of the block.

Hint

Frictional forces always act so as to oppose the direction of motion.
← Section 5.3

3. A particle of mass m kg is being pulled up a rough surface, which is inclined at 45° to the horizontal, by a force of 20 N, which acts parallel to the surface. The coefficient of friction between the particle and the surface is 0.2. Given that the particle accelerates at 2 m s^{-2}, find the value of m.

Hint

To find m write an equation of motion for the particle.

4. E/P A box of mass 0.3 kg is held at rest on a rough surface that is inclined at 30° to the horizontal. The coefficient of friction between the box and the surface is 0.15. The box is released from rest. Work out the time taken for the box to travel 5 m. (6 marks)

5. E/P A particle of mass 4 kg, initially at rest, is pulled up a rough surface, which is inclined at 25° to the horizontal, by a force of 50 N that acts at an angle of 30° to the surface.
The coefficient of friction between the particle and the surface is 0.1.

a. Work out the acceleration of the particle. (6 marks)

After 3 seconds the force is removed.

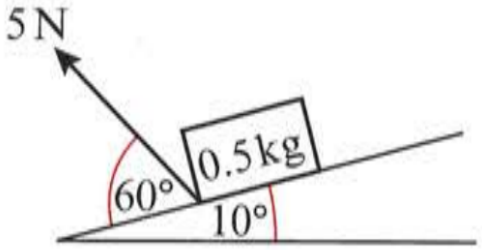
b. Work out the time taken for the particle to come to rest. (8 marks)

c. Show clearly whether the particle will remain at rest or begin to slide back down the slope. (8 marks)

6. E/P A particle of mass 4 kg is released from rest on a rough slope that is inclined at 35° to the horizontal. After 3 seconds the particle has travelled 4 m. Work out the coefficient of friction between the particle and the slope. (8 marks)

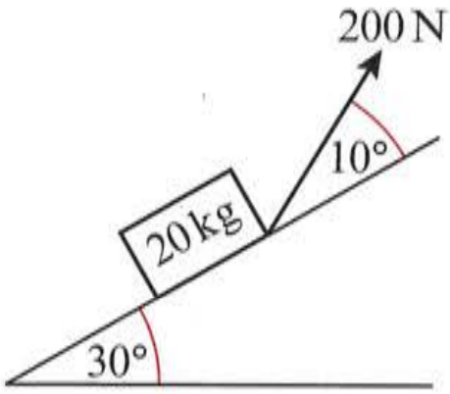
7. E/P A particle of mass 2 kg is moving up a rough slope inclined at 5° to the horizontal with constant acceleration. After 2 seconds the particle has a velocity of 4 m s^{-1}. After 6 seconds the particle has come to instantaneous rest. (6 marks)
Work out the coefficient of friction between the particle and the slope.

8. E/P A particle of mass 0.5 kg, initially at rest, is pulled down a rough surface, which is inclined at 10° to the horizontal, by a force of 5 N that acts at an angle of 60° to the surface. The coefficient of friction between the particle and the surface is 0.75.



Find the time taken for the velocity of the particle to reach 30 m s^{-1}. (6 marks)

9. E/P A sled of mass 20 kg, initially at rest at point A, is being pulled by a rope up a rough plane, which is inclined at 30° to the horizontal, to point B. The tension in the rope is constant, equal to 200 N, and acts at an angle of 10° to the plane. The coefficient of friction between the sled and the plane is 0.3. After 10 seconds of pulling, the rope snaps when the sled is 60 m from B.



Determine whether the sled reaches point B. You must justify your answer. (12 marks)

7.6 Connected particles

Practice Book, printed page 63.

1. A car of mass 1000 kg is towing a trailer of mass 200 kg up a straight hill that is angled at 10° to the horizontal. The car engine exerts a constant force of 10 kN. The car and the trailer are connected by a tow-bar. The total resistances to motion of the car and the trailer are modelled as being constant and of magnitudes 200 N and 150 N respectively.

a. Draw a force diagram to represent this situation.

b. By modelling the car and the trailer as particles and the tow-bar as a light inextensible rod, work out:

i. the acceleration of the car and trailer

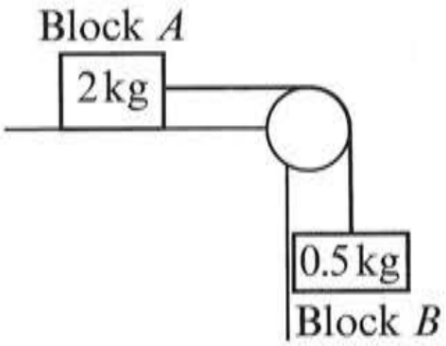
ii. the tension in the tow-bar.

c. State how you have used the fact that the tow-bar is inextensible in your calculations.

Hint

To work out the acceleration of the car and trailer, consider them together.

2. A block A of mass 2 kg rests on a rough horizontal table and is attached to one end of a light inextensible string. The string passes over a small smooth pulley fixed at the edge of the table. The other end of the string is attached to a block B of mass 0.5 kg, which hangs freely below the pulley. The coefficient of friction between block A and the table is 0.2.



The system is released from rest with the string taut.

a. Copy and complete the diagram to show the forces acting on the blocks.

b. Find the acceleration of block A.

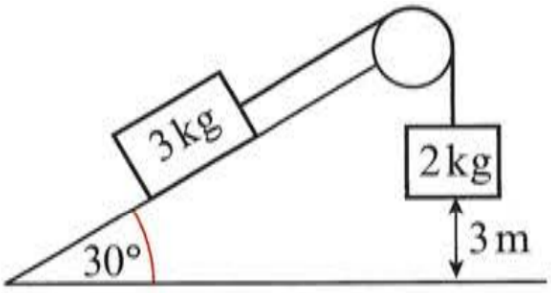
c. Work out the tension in the string.

d. State how you have used the fact that the pulley is smooth in your calculations.

Hint

To work out the acceleration, write equations of motion for the two blocks, then solve the equations simultaneously.

3. A 3 kg mass lies on a smooth plane that is inclined at 30° to the horizontal. It is attached to a 2 kg mass that hangs freely at a height of 3 m above the ground from a light inextensible string that passes over a small smooth pulley.



The system is released from rest with the string taut, and in the subsequent motion the 3 kg mass does not reach the pulley.

a. Work out the tension in the string when the system is released.

b. Find the acceleration of the system.

c. Find the time taken for the 2 kg mass to reach the ground.

Hint

You can work out either the acceleration or the tension first.

4. E/P A van of mass 1200 kg is towing a trailer of mass 250 kg up a straight hill that is inclined at 12° to the horizontal. The van engine exerts a constant force of 12 kN and the van and trailer are accelerating at 5 m s^{-2}. The resistances to motion of the van and trailer are modelled as being constant.

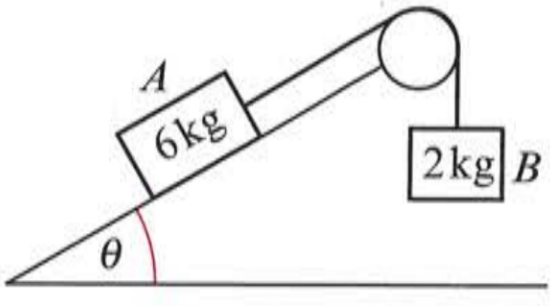
a. Work out the total resistances to motion of the van and trailer. (3 marks)

b. Given that the resistance to motion of the van is three times the resistance to motion of the trailer, work out the tension in the tow-bar. (2 marks)

When the van and trailer are travelling at 8 m s^{-1}, the tow-bar snaps.

c. Assuming that the resistance to motion remains unchanged, work out the time taken for the trailer to come to rest. (4 marks)

5. E/P Box A of mass 6 kg rests on a rough plane which is inclined at an angle θ to the horizontal, where $\tan\theta = 0.75$. The coefficient of friction between box A and the plane is μ. Box A is attached to one end of a light inextensible string, which passes over a smooth pulley. The string is attached to a second box, B, of mass 2 kg, which hangs freely below the pulley, as shown in the diagram.



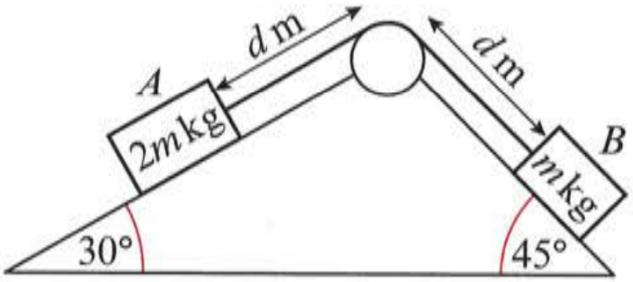
The system is released from rest.

a. Given that box A begins to slide down the slope, find the range of possible values of μ. (3 marks)

b. Given further that $\mu = 0.2$, find the time taken for box A to reach a speed of 5 m s^{-1}. (4 marks)

c. State two assumptions you have made in order to answer part **b**. You may not include any modelling assumptions given in the question. (2 marks)

6. E/P Block A of mass $2m$ kg and block B of mass m kg sit on smooth inclined planes angled at 30° and 45° to the horizontal respectively. The blocks are attached at either end of a light inextensible string. The string passes over a small smooth pulley at the top of the planes. Both blocks are initially at rest and at a distance of d m from the pulley.



The blocks are released from rest.

a. Show that block B begins to slide up the slope. (3 marks)

b. Show that the time taken for block B to hit the pulley is $\sqrt{\dfrac{12d}{(2-\sqrt{2})g}}$ (9 marks)

Problem solving: Set A

Bronze

A particle of mass 6 kg is at rest on a rough slope that is inclined at 15° to the horizontal. The particle is projected up the slope with initial speed 10 m s^{-1}. The coefficient of friction between the particle and the slope is 0.35.

a. Find the magnitude of the frictional force acting on the particle. (3 marks)

b. Work out the time taken for the particle to come to rest. (2 marks)

c. State how your answer to part **b** would change if the coefficient of friction between the particle and the slope were less than 0.35. (1 mark)

Silver

A particle of mass 2 kg is at rest on a rough slope that is inclined at an angle α to the horizontal. The particle is projected up the slope with initial speed 20 m s^{-1}. The coefficient of friction between the particle and the slope is 0.2. Given that the particle comes to rest in limiting equilibrium, find:

a. the size of angle α (3 marks)

b. the time taken for the particle to come to rest. (4 marks)

c. State, with a reason, how the size of α would differ if the coefficient of friction between the particle and the slope were greater than 0.2. (2 marks)

Gold

A particle of mass 3 kg is at rest on a rough slope that is inclined at α to the horizontal. The particle is projected up the slope with initial speed 15 m s^{-1}. The coefficient of friction between the particle and the slope is 0.15.

a. Given that the particle comes to instantaneous rest and then immediately starts to slide back down the slope, find the range of possible sizes of α. (3 marks)

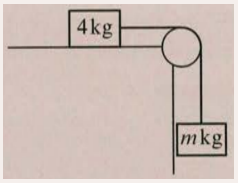
b. Given further that $\alpha = 20°$, find the total time taken from the time the particle is projected to when it returns to its starting position. (7 marks)

c. State, with reasons, what effect increasing the mass of the particle would have on your answers to parts **a** and **b**. (2 marks)

Problem solving: Set B

Bronze

A box of mass 4 kg rests on a rough horizontal table and is attached to one end of a light inextensible string. The string passes over a small smooth pulley fixed at the edge of the table. The other end of the string is attached to a box of mass m kg which hangs freely below the pulley. The coefficient of friction between the box and the table is 0.375. The system is released from rest with the string taut, causing the boxes to accelerate at 1.2 m s^{-2}.



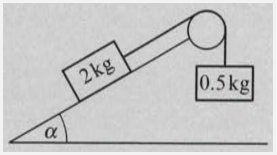
a. Work out the tension in the string. (3 marks)

b. Work out the value of m. (2 marks)

c. State how you have used the fact that the pulley is smooth in your calculations. (1 mark)

Silver

A block of mass 2 kg rests on a smooth slope that is inclined at an angle α to the horizontal. It is attached to one end of a light inextensible string. The string passes over a small smooth pulley fixed at the top of the slope. The other end of the string is attached to a mass of 0.5 kg which hangs freely below the pulley. The system is released from rest with the string taut, causing the boxes to accelerate at 0.8 m s^{-2}.



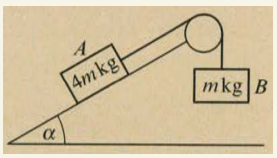
a. Work out the two possible sizes of angle α. (5 marks)

b. Given that α takes the lesser of these two sizes, find the magnitude of the force exerted on the pulley by the string. (4 marks)

c. State how your answer to part **b** would differ if α took the greater of its two possible sizes. (1 mark)

Gold

Block A of mass $4m$ kg is held at rest on a rough slope that is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{5}{12}$. It is attached to one end of a light inextensible string. The string passes over a small smooth pulley fixed at the top of the slope. The other end of the string is attached to a block B of mass m kg which hangs freely below the pulley. The coefficient of friction between block A and the slope is μ. The system is released from rest. Given that block A begins to slide down the slope,



a. find the range of possible values of μ. (4 marks)

b. Given that $\mu = 0.1$, find the acceleration of the system. (8 marks)

c. State, with reasons, the effect that doubling m would have on:

i. the acceleration of the system

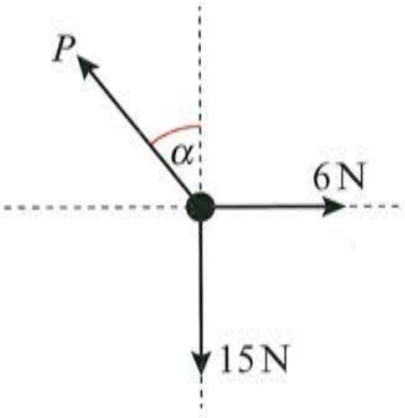
ii. the tension in the string.

(3 marks)

Selected exam question bank questions

Question 26

A particle of mass 3 kg is held in equilibrium by three forces as shown in the diagram.



a. Find the size of angle α. (3 marks)

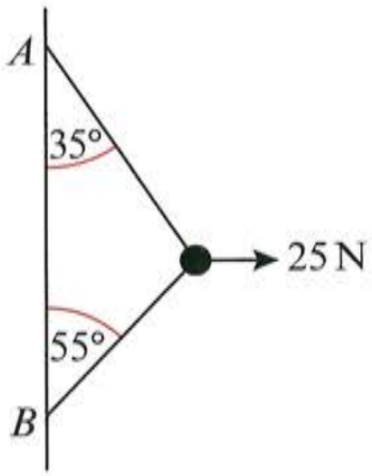
b. Calculate the magnitude of P. (1 mark)

The force P is removed.

c. Find the time taken for the particle to travel a distance of 10 m. (4 marks)

Question 34

A bead is threaded on a string that is attached to a vertical wall at the points A and B. The bead is held in equilibrium by a horizontal force of magnitude 25 N. The string is modelled as being light and inextensible, and the bead is modelled as being smooth.



a. Find the tension in the string. (2 marks)

b. Work out the mass of the bead. (2 marks)

c. State how you have used the fact that the bead is smooth in your calculations. (1 mark)

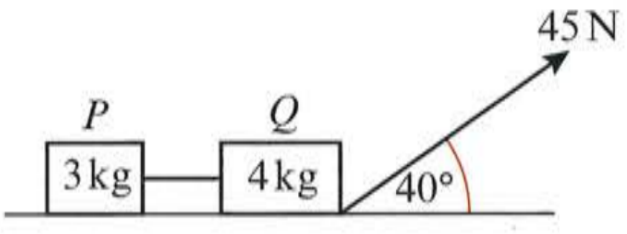
d. Briefly explain how the tension in the string would be affected if the string is not modelled as being light. (1 mark)

Question 40

A uniform ladder AB has mass m kg and length 2 m. End A rests on rough horizontal ground and end B rests against a smooth vertical wall. The ladder makes an angle of $60°$ with the horizontal. A man of mass $3m$ kg stands on the ladder at C where $AC = 0.5$ m. Given that the ladder is in limiting equilibrium, work out value of the coefficient of friction between the ladder and the ground. (8 marks)

Question 43

Two boxes P and Q, of masses 3 kg and 4 kg respectively, are joined by a light inextensible string. Initially the boxes are at rest on a rough horizontal plane with the string taut. A constant force of 45 N is applied to Q at an angle of $40°$ as shown in the diagram.



The force is applied for 3 seconds and during this time Q travels a distance of 5 m. The coefficient of friction between P and the plane is μ and the coefficient of friction between Q and the plane is 2μ.

a. Work out the acceleration of Q. (2 marks)

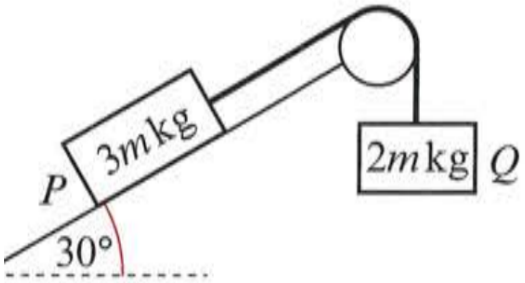
b. Calculate the value of μ. (4 marks)

c. Find the tension in the string. (3 marks)

d. State how you have used the fact that the string is inextensible in your calculations. (1 mark)

Question 46

Two boxes P and Q, of masses $3m$ kg and $2m$ kg respectively, are attached to the ends of a light inextensible string. Box P lies on a rough surface that is inclined at 30° to the horizontal. The coefficient of friction between P and the surface is 0.1.



The string passes over a small smooth pulley fixed on the edge of the surface. Box Q hangs freely below the pulley.

The system is released from rest with the strings taut.

a. Work out the time taken for P to reach a speed of 10 m s^{-1}. (5 marks)

b. State one assumption you have made in your calculation for part a. (1 mark)

At the instant P has velocity 10 m s^{-1}, the string snaps.

c. Work out the distance travelled by P from the time the string snaps to the time P comes to instantaneous rest. (5 marks)

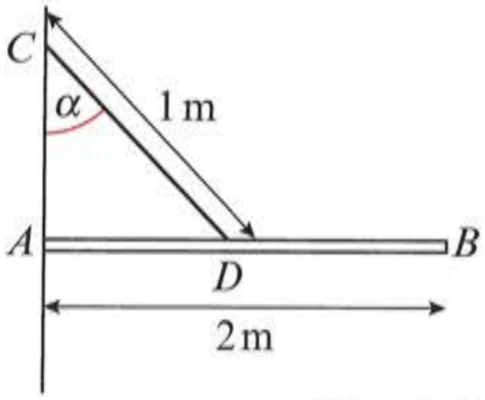
d. Determine whether P will remain at rest or begin to slide back down the slope, fully justifying your answer. (2 marks)

Question 49

A non-uniform ladder AB, of mass m and length l, rests in equilibrium with the end A on a rough horizontal floor and the end B against a rough vertical wall. The ladder makes an angle of 60° with the floor. The coefficient of friction between the ladder and the floor is 0.3 and the coefficient of friction between the ladder and the wall is 0.7. The centre of mass of the ladder is at the point M. Given that the ladder is on the point of slipping at both ends, determine, in terms of l, the distance AM. (12 marks)

Question 52

A beam AB, of weight 20 N and length 2 m, is smoothly hinged to a vertical wall at A. The beam is held in equilibrium in a horizontal position by a rope of length 1 m. One end of the rope is fixed to the point C vertically above A and the other end is fixed at a point D such that the angle ACD is α. The beam is modelled as a uniform rod and the rope is modelled as a light inextensible string.



a. Show that the tension in the string is equal to $\dfrac{40}{\sin 2\alpha}$ (4 marks)

b. Given that the tension in the string is 80 N, find the two possible sizes of angle α. (3 marks)

Given further that α takes the lesser of these two values,

c. find the direction and magnitude of the force exerted on the beam by the hinge at A. (8 marks)

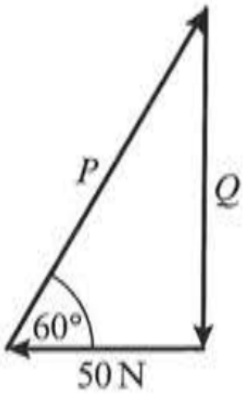
d. Without further calculation, state how the direction of this force would change if α took the greater of its two possible values. (2 marks)

Answers

7.1 Static particles

1.
- **a.** $P\cos 60° + Q\cos 45° = 40$
- **b.** $P\sin 60° = Q\sin 45°$
- **c.** $P = 29.3$ N (3 s.f.), $Q = 35.9$ N (3 s.f.)

2.
- **a.**



- **b.** $P = 100$ N, $Q = 50\sqrt{3}$ N

3. $P = 5$ N, $Q = 5\sqrt{3}$ N

4. $P = 10$ N, $Q = 10\sqrt{3}$ N

5. $P = 22.2$ N (3 s.f.), $Q = 44.1$ N (3 s.f.)

6.
- **a.** $\alpha = 41.8°$ (3 s.f.)
- **b.** 22.4 N (3 s.f.)

7.2 Modelling with statics

1. $2\sqrt{3}$ N

2.
- **a.** $15\sqrt{2}$ N
- **b.** $15(1 + \sqrt{2})$ N
- **c.** Tension is the same on both sides of the bead.

3.
- **a.** $\frac{60g}{13}$ N
- **b.** $\frac{25g}{13}$

4. $T_P = 14.3$ N (3 s.f.), $T_Q = 10.1$ N (3 s.f.)

5.
- **a.** 11.4 N (3 s.f.)
- **b.** 0.143 kg (3 s.f.)
- **c.** The tension in the string would be greater at the top of the string as it would have to support the additional weight of the string below it.

6.
- **a.** 45.6° (3 s.f.)
- **b.** The angle would be smaller.

7. R($\nearrow$) : $2g - mg\sin\alpha = 0$, rearrange to find $\alpha = \arcsin\left(\frac{2}{m}\right)$

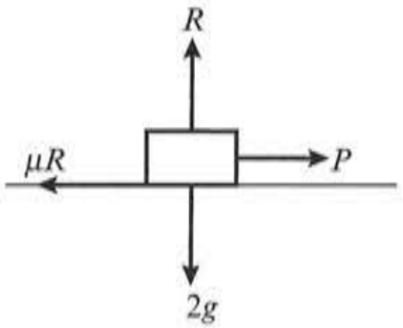
8. R($\nearrow$) : $4P\cos 30° - P\cos 60° - 20\sin 30° = 0$ so

$$P = \frac{20\sin 30°}{4\cos 30° - \cos 60°} = 3.3737\ldots$$

R($\nwarrow$) : $R + 4P\sin 30° + P\sin 60° - 20\cos 30° = 0$
$R = 20\cos 30° - P(4\sin 30° + \sin 60°)$
Substituting gives $R = 7.65$ N as required.

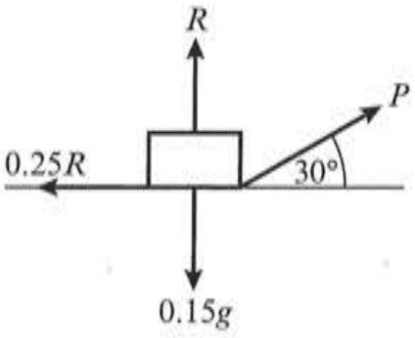
7.3 Friction and static particles

1. **a.**



b. 3.92 N

2. **a.**



b. 0.371 N (3 s.f.)

3. **a.** 37.9 N (3 s.f.)
b. 0.268 (3 s.f.)

4. **a.** 23.3 N (3 s.f.)
b. 14.6 N (3 s.f.)

5. 0.603 or 3.32 (3 s.f.)

6. R($\nwarrow$) : $R = mg\cos\alpha$ R($\nearrow$) : $\mu R = g + mg\sin\alpha$
Substitute and rearrange to find μ.

7. **a.** 702 (3 s.f.)
b. Valid because air resistance not a factor in static situation; rotational forces unlikely to affect situation.

8. $0.116 \leq F \leq 0.549$ (3 s.f.)

7.4 Static rigid bodies

1.

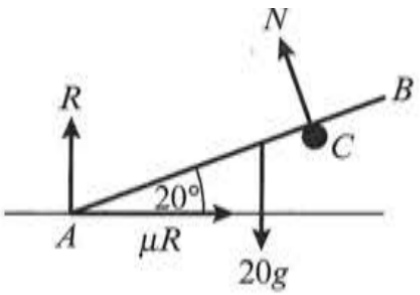
a. $N = F$

b. $R = 25g$ N

c. 70.7 N (3 s.f.)

2.

a.



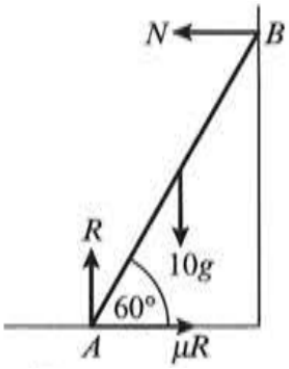
b. 132 N (3 s.f.)

c. 0.621

d. Reaction at the peg acts perpendicular to the rod.

3.

a.



b. $\frac{5\sqrt{3}g}{3}$ N

c. $\frac{\sqrt{3}}{6}$

4.

a. 63.4° (3 s.f.)

b. Assumption means no friction at points of contact with the ladder. Wall and floor are unlikely to be smooth in real life.

c. α would be smaller.

5.

a. 282 N (3 s.f.)

b. 64.4° (3 s.f.)

6.

a. 107 N (3 s.f.)

b. 165 N (3 s.f.)

7.

a. 539 N

b. 483 N (3 s.f.) at an angle of 14.7° (3 s.f.) below the horizontal.

8.

a. Take moments about A to find $T = \frac{2Mg}{3\sin\alpha}$

b. R($\rightarrow$) :

$$R_x = T\cos\alpha = \frac{2Mg}{3\sin\alpha}\cos\alpha = \frac{2Mg}{3\tan\alpha}$$

R($\uparrow$) :

$$R_y + T\sin\alpha = Mg$$

$$R_y + \frac{2Mg}{3\sin\alpha}\sin\alpha = Mg$$

$$R_y + \frac{2}{3}Mg = Mg$$

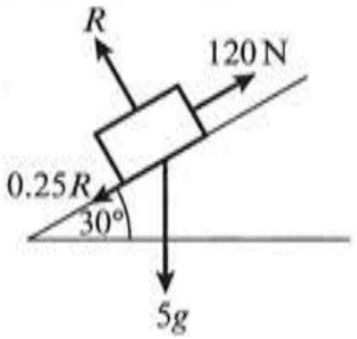
$$R_y = \frac{1}{3}Mg$$

c. 38.7°

d. The angle would be larger (because $\tan\alpha = 2\mu$).

7.5 Dynamics and inclined planes

1. **a.**



b. 42.4 N

c. 17.0 m s^{-2} (3 s.f.)

2.

a. 1.96 N up the slope

b. R($\swarrow$) to find the resultant force acting on the block, which is 6.53 N (3 s.f.) down the slope, therefore the block will begin to slide.

c. 6.53 m s^{-2} (3 s.f.)

3. 1.94 (3 s.f.)

4. 1.66 seconds (3 s.f.)

5.

a. 6.42 m s^{-2} (3 s.f.)

b. 5.92 seconds (3 s.f.)

c. R($\swarrow$) to find the resultant force acting on the particle to be 13.0 N (3 s.f.) (acting down the slope), therefore the particle will begin to slide.

6. 0.589 (3 s.f.)

7. 0.0149 (3 s.f.)

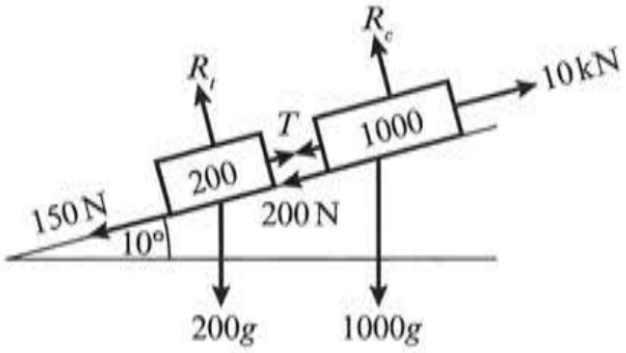
8. 5.03 seconds

9. When the rope snaps the sled has velocity 29.2 m s^{-1} (3 s.f.) and acceleration of -7.45 m s^{-2} (3 s.f.) up the slope.
Use $v^2 = u^2 + 2as$ to find that the sled will travel 57.3 m (3 s.f.) before coming to rest. Therefore the sled will not reach B.

7.6 Connected particles

1.

a.



b.

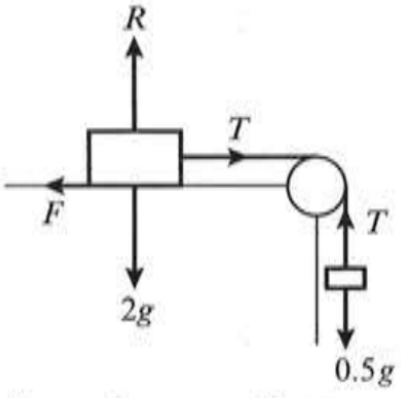
i. 6.34 m s^{-2} (3 s.f.)

ii. 1760 N (3 s.f.)

c. Acceleration of car and trailer are the same.

2.

a.



b. $\frac{1}{25}g$ m s^{-2}

c. $\frac{12}{25}g$ N

d. Tension in string is the same on both sides of the pulley.

3.

a. $1.8g$ N

b. $0.1g$ m s^{-2}

c. 2.47 seconds

4.

a. 1800 N (3 s.f.)

b. 2210 N (3 s.f.)

c. 2.09 seconds (3 s.f.)

5.

a. $\mu < \frac{1}{3}$

b. 6.38 seconds

c. For example, box A does not reach ground; box B does not reach pulley; boxes can be modelled as particles.

6.

a. $2mg\sin 30° = mg$
$mg\sin 45° = \frac{1}{\sqrt{2}}mg < mg$
So block B slides up the slope.

b. Resolve parallel to the plane for each block to find $T = mg - 2ma$ and $T = ma + \frac{\sqrt{2}}{2}mg$
Then solve to find $a = \frac{2-\sqrt{2}}{6}g$
Use $s = ut + \frac{1}{2}at^2$ with $u = 0$, $s = d$, $a = \frac{2-\sqrt{2}}{6}g$ to find t.

Problem solving: Set A

Bronze

a. 19.9 N (3 s.f.)
b. 1.71 seconds
c. It would take longer to come to rest.

Silver

a. 11.3° (3 s.f.)
b. 5.21 seconds (3 s.f.)
c. The size of α would be greater, as a greater component of the weight of the particle would need to act down the slope to maintain limiting equilibrium.

Gold

a. $\alpha > 8.53°$ (3 s.f.)
b. 6.34 seconds (3 s.f.)
c. The component of the weight that acts down the slope and the frictional force both have a factor of m, so the conditions for limiting equilibrium and the acceleration of the particle would be unchanged.

Problem solving: Set B

Bronze

a. 19.5 N
b. 2.27 (3 s.f.)
c. Tension equal on both sides of the pulley.

Silver

a. 8.51° (3 s.f.) or 20.6° (3 s.f.)
b. 6.82 N (3 s.f.)
c. The force exerted on the pulley would be greater.

Gold

a. $\mu < \frac{7}{48}$
b. $a = \frac{11}{325}g$
c.
  i. a is independent of m, so the acceleration would be unchanged.
  ii. Tension initially is $T = mg + ma$; when m is doubled $T = 2mg + 2ma = 2(mg + ma)$, so the tension in the string would double.

Selected exam question bank answers

Question 26

a. 21.8° (3 s.f.)
b. 16.2 N (3 s.f.)
c. 1.93 seconds

Question 34

a. 18.0 N (3 s.f.)
b. 0.450 kg (3 s.f.)
c. The tension in the string is the same on both sides of the bead.
d. The tension in the string would be greater at the top as it would have to support the mass of the string below it.

Question 40 0.180 (3 s.f.)

Question 43

a. $\frac{10}{9}$ m s^{-2}
b. 0.534 (3 s.f.)
c. 19.0 N (3 s.f.)
d. Both particles have the same acceleration.

Question 46

a. 2.12 seconds
b. P does not reach the pulley, Q does not reach the floor.
c. 8.70 m (3 s.f.)
d. R($\swarrow$) : Force $= 3mg\sin 30° = 1.5mg$

$$F_{\max} = \mu_s R = 0.1 \times 3mg\cos 30° = \frac{3\sqrt{3}}{20}mg < 1.5mg$$

So P slides back down the slope.

Question 49 $0.392l$ (3 s.f.)

Question 52

a. $AD = \sin\alpha$
Moments about A: $T\sin\alpha\cos\alpha = 20$
So $T = \frac{20}{\frac{1}{2}\sin 2\alpha} = \frac{40}{\sin 2\alpha}$
b. 15° and 75°
c. 60.9 N at an angle of 70.1° to the horizontal below the beam.
d. The angle would be closer to the horizontal (or above the beam).